

OLARINDE OLUBUNMI RACHEAL

Austin TX | 479-250-7266 | onadipeolubunmi@gmail.com | [LinkedIn](#) | [Portfolio](#)

ABOUT ME

Versatile Data Scientist and Machine Learning Engineer with a solid foundation in data analytics, statistical modeling, and AI systems development. Passionate about building data-driven solutions that are not only technically sound but also fair, human-centered, and socially responsible. Skills set spans machine learning, deep learning, business intelligence, and data engineering making me well-suited for roles in applied AI, research, product development, and mission-driven innovation. With a multidisciplinary academic background and a unique mix of analytical rigor, creativity, and lived experience, I bring both technical depth and user-centered insight to every problem I tackle.

TECHNICAL SKILLS

Programming: Python, R, SQL, HTML, CSS

Cloud/DevOps: AWS (S3, SageMaker, Lambda, EC2), Docker (basic), Git, GitHub

Machine Learning & Data Science: Supervised & Unsupervised Learning, Statistical Modeling, Feature Engineering, A/B Testing, EDA

Data Visualization & Business Intelligence:

Tableau, Pandas, Matplotlib, Seaborn, Plotly

Deep Learning & AI: Neural Networks, CNNs, Transfer Learning, Computer Vision, NLP

Frameworks: Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face (Transformers), FAISS, OpenCV, spaCy, NLTK, Gensim

EXPERIENCE

Smart Sickle Cell Diary App (Independent)

Founder & Lead ML Engineer - Smart Sickle Cell Diary App | June 2025 - November 2025

- Designed and deployed a Streamlit app (Hugging Face Spaces) for people living with SCD to log symptoms and receive LLM-powered insights with trigger detection (cold, dehydration, stress, exertion).
- Built a lightweight RAG pipeline (FAISS) and Insight Card UX; added tester feedback loop and CSV export for personal records.
- Ensured ethical AI, user agency, and fairness by implementing transparent algorithms and user control features, fostering trust and inclusivity in the app
- Ran a 14-day mini-pilot with de-identified logs; improved onboarding via a how-to video and in-app tips
- Utilized Python, pandas, FAISS, Streamlit, OpenAI, Hugging Face, and GitHub Pages to develop the app, ensuring robust functionality and seamless integration of technologies
- Developed and launched a portfolio site for the Smart Sickle Cell Diary App using HTML and Streamlit, enhancing user engagement and accessibility

Primrose School at Rancho Sienna, Georgetown TX*Teacher Support | Oct 2020 - Feb 2025*

- Designed and delivered data-informed lesson plans, applying structured workflows analogous to data pipelines.
- Developed strong communication skills applicable to technical documentation and stakeholder alignment.
- Supported diverse learners, experience that translates to user-centered AI system design

Mustard Seed Children Academy, Bentonville AR*Teacher | Sept 2018 - Apr 2019*

- Used student performance data to refine instruction and improve outcomes
- Built collaboration and communication skills transferable to agile tech teams

Nigeria Cleaning Service Limited, Lagos Nigeria*Accounts Clerk | May 2008 - Dec 2008*

- Maintained ledgers and processed invoices with high accuracy
- Improved billing cycles and developed documentation skills relevant to data management

PROJECTS<https://github.com/Olunmi-Amoke>**Smart Sickle Cell Diary App (Capstone)***Streamlit App | NLP + ML | Ethical AI*

- Built an interactive diary app for people living with SCD, combining free-text + light structured inputs and trend visualization for reflection.
- Trained ML models to predict pain intensity (regression) and emotion (classification); exported models with Joblib and integrated into Streamlit.
- Emphasized ethical AI, user agency, and fairness.
- **Tools:** Streamlit, Scikit-Learn, TextBlob, Plotly, Python, Joblib

Sentiment Analysis of Customer Reviews*NLP | Transformers | Embedding Comparison*

- Built and evaluated sentiment classifiers using Word2Vec, Glove, and Sentence Transformers.
- Fine-tuned a transformer from Hugging Face to summarize reviews and extract impactful weekly events.
- **Tools:** Python, Scikit-Learn, Gensim, Hugging Face Transformers, Matplotlib, Pandas

Landmark Classification Project Using CNNs*Computer Vision | Transfer Learning*

- Improved model accuracy from 41% to 79% using ResNet50 with fine-tuning and image augmentation.
- Developed modular PyTorch training scripts and visualized training/validation metrics.
- **Tools:** PyTorch, OpenCV, Pillow, Livelossplot

Scones Unlimited – ML Workflow with AWS*End-to-End ML Deployment | AWS*

- Designed a serverless ML pipeline for image classification using AWS SageMaker, Lambda, and S3.
- Deployed and monitored models with SageMaker Model Monitor to detect drift and performance issues.
- **Tools:** AWS (S3, SageMaker, Lambda), Boto3, Python

EDUCATION

Eastern University, St. Davids, PA

Master of Science in Data Science (Completed July 2025)

- **Completed Courses:** Data Manipulation, Data Science for Business, Data Analytics in R, Data and Database Management with SQL, Cloud Foundations, Fundamentals of Machine Learning Models, Applied Machine Learning, Natural Language Processing, Ethical and Philosophical Issues in Data Science, Applied Data Science & Analytics
- **Capstone Projects:**
 - “*The Model Doesn’t Speak My Pain*” – Investigated algorithmic bias in symptom interpretation for marginalized health communities using NLP and ethics frameworks.
 - *Smart Sickle Diary App* - Developed a machine learning app to analyze patient-reported pain/emotion logs and visualize insights using NLP and Streamlit.

University Of Texas at Austin, TX

Postgraduate Certificate in AI/ML (2024–2025)

Obafemi Awolowo University, Ile-Ife, Nigeria

Bachelor of Arts in Music

The Polytechnic Ibadan, Ibadan, Nigeria

Associate’s Degree in Music Technology

CERTIFICATIONS

- **Udacity Nanodegree Programs 2024**
AWS Machine Learning Engineer | AI Programming with Python | ML Fundamentals
- **Codecademy 2024**
ML/AI Engineer | NLP for Data Scientists | Machine Learning for Data Scientists
- **AWS & Udacity AI/ML Scholar 2023, 2024**
Selected among top applicants for hands-on, project-based learning in cloud ML deployment, deep learning, and AI automation.